All metrics come from the four cells of the confusion matrix

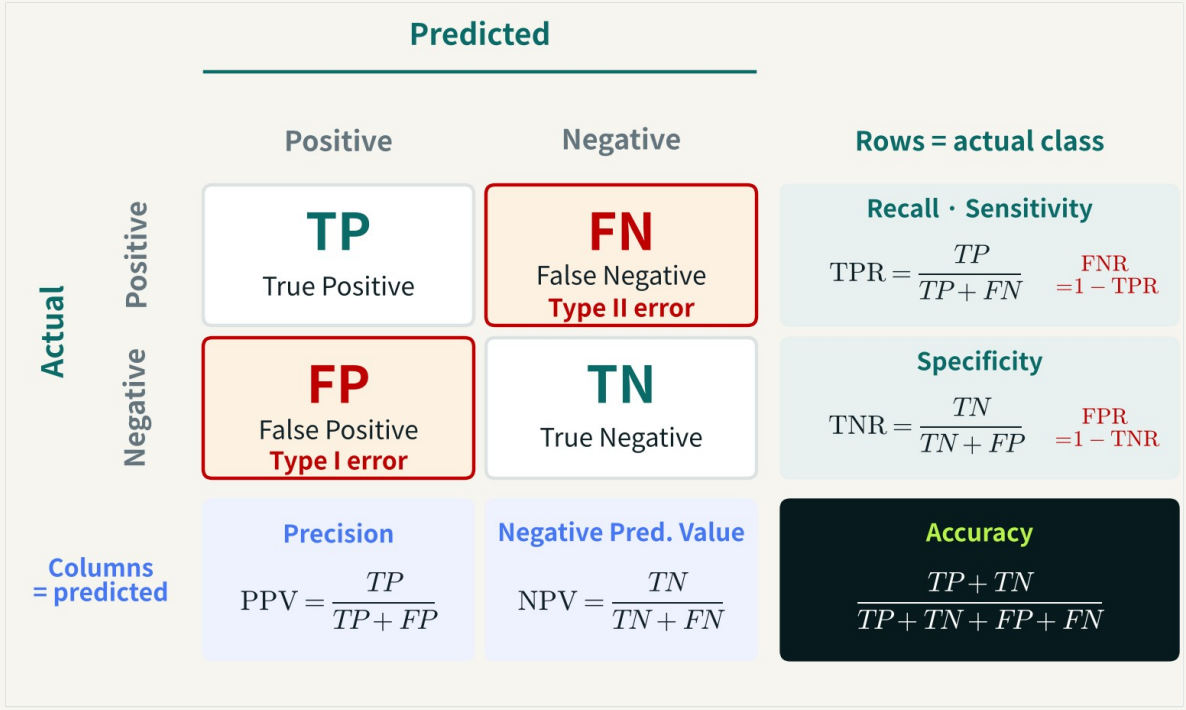
scikit-learn convention: rows are actual labels, columns are predicted labels

How to read it

- Rows = actual class
- Columns = predicted
- Diagonal = correct

※ Axes differ by source

FP = Type I error
FN = Type II error



Recall penalizes missed positives (FN); Precision penalizes false alarms (FP)

Both extremes fail: misses and false alarms must be weighed together

Two extreme models applied to 1,000 transactions containing 10 frauds

① Predict all 'normal' KPI: accuracy

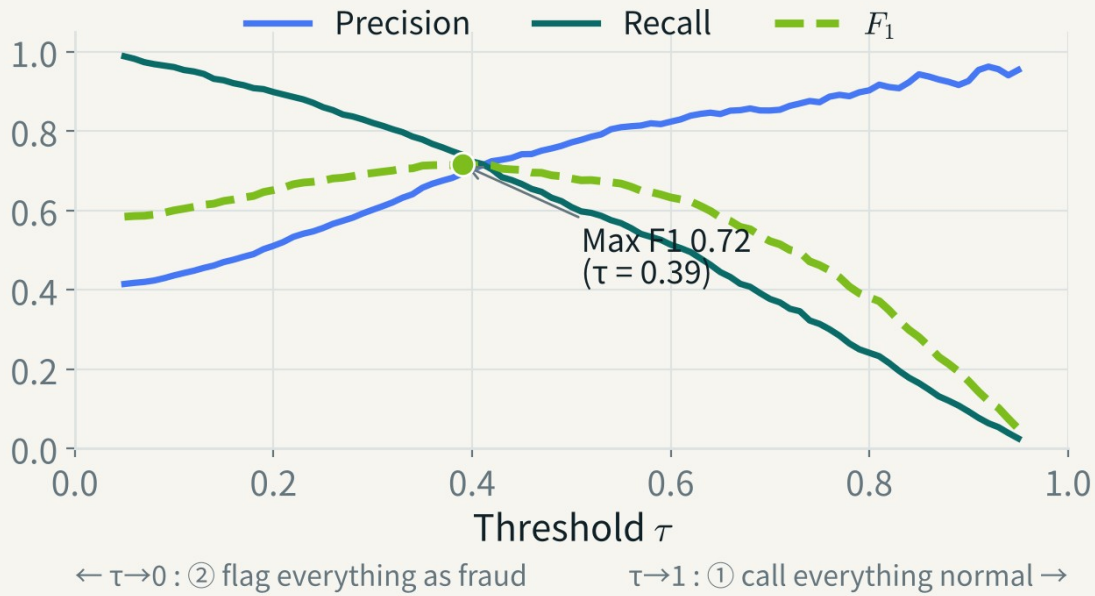
Misses all 10 frauds: missing (FN)

$$\text{Accuracy} = \frac{990}{1000} = 99\%, \quad \text{Recall} = \frac{0}{10} = 0, \quad F_1 = 0$$

② Predict all 'fraud' KPI: catch all fraud

Flags 990 honest customers: false alarm (FP)

$$\text{Recall} = \frac{10}{10} = 100\%, \quad \text{Precision} = \frac{10}{1000} = 1\%, \quad F_1 \approx 0.02$$



① looks near-perfect on Accuracy and ② on Recall, yet neither model is usable

F1 scores both extremes near zero, demanding a balance of misses and false alarms

F1 is high only when Precision and Recall are both high

The harmonic mean is pulled toward the smaller of the two values

Definition · the reciprocal of the mean of reciprocals

$$F_1 = \frac{2}{\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

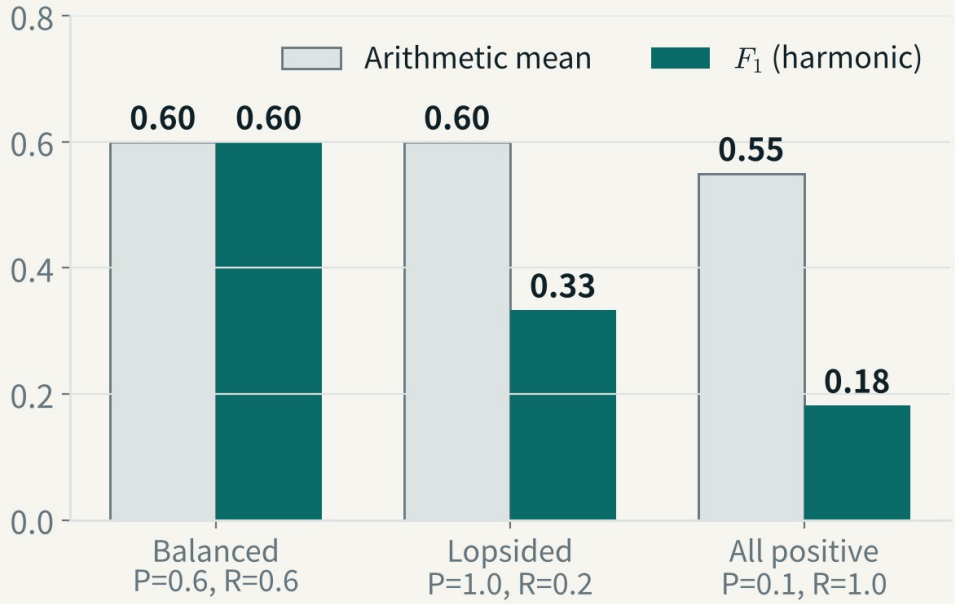
Worked example ① · TP = 40, FP = 10, FN = 20

$$\text{Precision} = \frac{40}{40+10} = 0.80, \qquad \text{Recall} = \frac{40}{40+20} \approx 0.67$$

Worked example ② · same value straight from the matrix

$$F_1 = \frac{2 \cdot 0.80 \cdot 0.67}{0.80 + 0.67} = \frac{2TP}{2TP + FP + FN} = \frac{80}{110} \approx 0.73$$

Same arithmetic mean, different F1



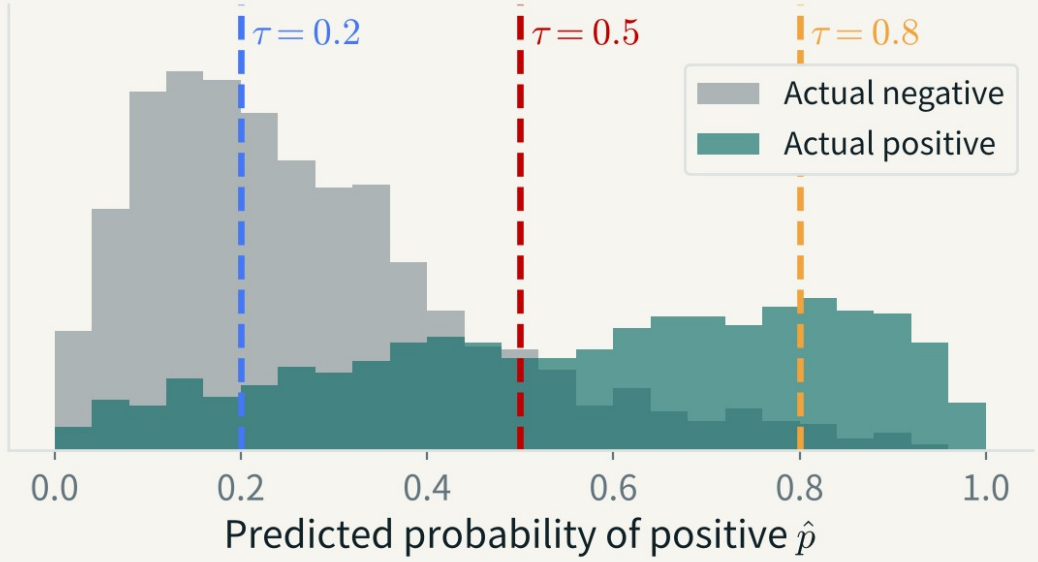
F1 ignores TN, so it suits rare-positive problems such as fraud detection and diagnosis

A lopsided model hides behind the arithmetic mean but is exposed by F1

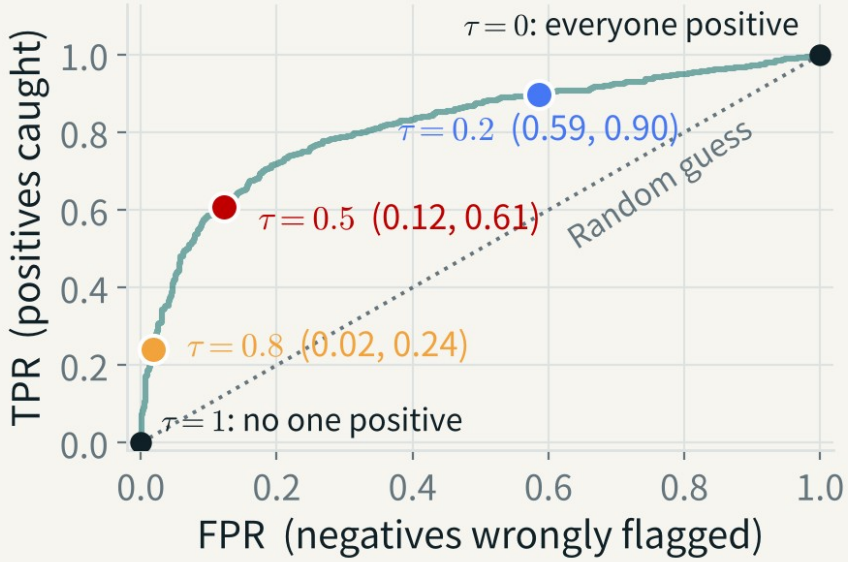
Connecting the points traced as the threshold moves gives the ROC curve

One threshold → one confusion matrix → one (FPR, TPR) point · logistic regression, synthetic data

① Everything right of τ is called positive



② Connect the dots → ROC (AUC = 0.82)



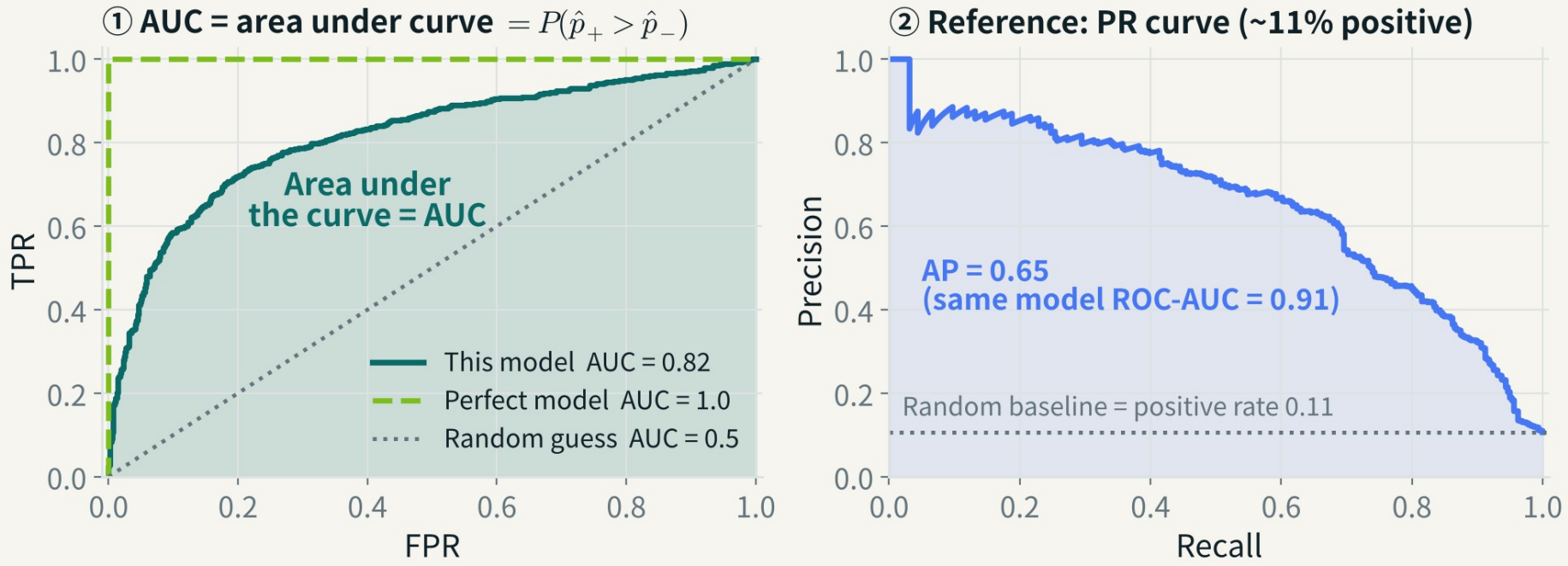
Lowering τ flags more positives, so TPR and FPR rise together and the point moves up and right

The closer the curve hugs the top-left, the better; the area under it is the AUC

※ AUC = probability a random positive scores higher than a random negative · 0.5 = coin flip, 1.0 = perfect

AUC summarizes performance across all thresholds in a single number

AUC = area under the ROC curve = chance a random positive outranks a random negative



AUC 0.82 means that in about 82% of positive–negative pairs the positive scores higher

When positives are rare, ROC-AUC is optimistic, so check the PR curve and AP too

※ Rule of thumb: ≥0.9 excellent · 0.8–0.9 good · 0.7–0.8 fair · 0.5 random / PR random baseline = positive rate